Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*	(i)	Capacitor equation rearranged $\frac{Cd}{\varepsilon_0} = A$ (1) Area = x^2 used (1) Correct answer = 42.5 [cm] (1)	1	1 1		3	3	
		(ii)	Logical method employed e.g. eliminating 1, 2 etc.(can be implied by correct calculation for 4) (1) Evidence of a correct capacitor in series calculation (1) Evidence of a correct capacitor in parallel calculation (1) Number 4 chosen (1)			4	4	2	
		(iii)	Dielectric / increase the permittivity Accept a named dielectric	1			1		
		(iv)	Valid method e.g. $\frac{q^2}{2C} = U$ or 2 other valid equations (1) Accept use of 3 equations Rearrangement or algebra (1) Answer = 9.96×10^{-8} [C] (accept 1×10^{-7} C) (1)	1	1 1		3	3	
	(b)	(i)	RC used as the time constant (1) Answer = 1.94 M[Ω] (1)	1	1		2	2	
		(ii)	2 time constants interpreted or substitution into: $q = q_0 e^{-t/CR}$ (1) 0.37^2 or rearrangement of equation **ecf** (1) Answer = 0.135 or 13.5% or $\frac{27}{200}$ (1)		3		3	3	
	(c)	(i)	Correct shape (starting from 0,0 increasing with decreasing gradient (1) 50-75% charged at 15.5 ms or 8-15 ms half-life (1)	1	1		2	2	2
		(ii)	Correct shape (starting from I_0, 0 decreasing and decreasing negative gradient) (1) 50-75% discharged at 15.5 ms or 8-15 ms half-life (1)	1	1		2	2	2
			Question 1 total	**6**	**10**	**4**	**20**	**17**	**4**